Heart Disease Prediction using ML Algorithms

Abstract

Heart disease is one of the leading causes of mortality worldwide. Early detection plays a crucial role in reducing the risk of severe cardiovascular complications and improving patient outcomes. This project presents a machine learning-based approach for predicting heart disease using clinical and demographic attributes. The study involves exploratory data analysis, data preprocessing, feature analysis, and the implementation of multiple machine learning algorithms.

Several classification models including Logistic Regression, Naive Bayes, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and XGBoost were evaluated. Performance was assessed using metrics such as accuracy, precision, recall, F1-score, and AUC-ROC.

Experimental results demonstrate that the Random Forest algorithm achieved the best performance with the highest accuracy and balanced evaluation metrics. The findings highlight the effectiveness of ensemble learning techniques in predicting heart disease and their potential application in assisting medical diagnosis.

1. Introduction

Heart disease is a major global health concern and one of the primary causes of death across many countries. According to global health statistics, cardiovascular diseases account for a significant proportion of mortality worldwide.

Early identification of individuals at risk of heart disease is essential for timely medical intervention and preventive healthcare strategies. However, traditional diagnostic approaches often rely heavily on manual clinical evaluation and may not fully leverage the predictive potential of available medical data.

Machine learning techniques offer powerful tools for analyzing healthcare datasets and identifying hidden patterns that can assist in medical diagnosis. By using predictive models trained on patient data, machine learning systems can support healthcare professionals in identifying high-risk individuals and improving diagnostic accuracy.

This project explores the use of multiple machine learning algorithms to predict heart disease based on clinical attributes and compares their performance to determine the most effective model.

2. Problem Statement

The increasing prevalence of heart diseases, a leading cause of mortality globally, necessitates effective measures for early detection and prevention. Despite significant advancements in medical science, many individuals remain deprived of timely diagnoses due to economic limitations, lack of awareness, or inadequate access to healthcare services.

This disparity highlights the urgent need for scalable and affordable diagnostic solutions. Given a dataset comprising features such as age, gender, blood pressure, cholesterol levels, blood sugar levels, and other relevant medical parameters, this study aims to develop a machine learning-based predictive system capable of identifying individuals at risk of heart disease.

The goal is to analyze these clinical attributes and build predictive models that can assist healthcare professionals in early diagnosis, risk assessment, and preventive intervention strategies.

3. Objectives

The primary aim of this study is to leverage machine learning algorithms to develop a predictive model for heart disease.

The key objectives include:

- Perform Exploratory Data Analysis (EDA) to identify patterns and relationships within the dataset.
- Conduct data preprocessing and feature analysis to prepare the dataset for modeling.
- Implement multiple machine learning classification algorithms.
- Evaluate the models using performance metrics such as accuracy, precision, recall, F1-score, and AUC-ROC.
- Identify the best-performing algorithm for heart disease prediction.
- Provide insights that may assist in improving healthcare risk assessment systems.

4. Dataset Description

The dataset used in this project contains various medical and demographic attributes related to heart disease prediction. These features represent clinical measurements and patient information used to determine the likelihood of heart disease.

Dataset Attributes

age
Age of the individual (in years)

gender
Gender of the individual (male or female)

chest_pain_type
Type of chest pain experienced by the patient

resting_blood_pressure
Resting blood pressure measured in mm Hg

serum_cholesterol
Serum cholesterol level measured in mg/dl

fasting_blood_sugar
Fasting blood sugar (1 if >120 mg/dl, otherwise 0)

resting_ecg
Resting electrocardiographic results

max_heart_rate_achieved
Maximum heart rate achieved during exercise

exercise_induced_angina
Exercise induced angina (1 = yes, 0 = no)

oldpeak
ST depression induced by exercise relative to rest

slope_of_peak_exercise_st_segment
Slope of the peak exercise ST segment

number_of_vessels_fluro
Number of major vessels colored by fluoroscopy

thal
Type of thalassemia defect

target
Target variable indicating the presence of heart disease (1 = disease present, 0 = no disease)

5. Data Preprocessing

Before applying machine learning models, the dataset underwent several preprocessing steps to ensure data quality and reliability.

Dataset Overview

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       303 non-null    int64
 1   sex       303 non-null    int64
 2   cp        303 non-null    int64
 3   trestbps  303 non-null    int64
 4   chol      303 non-null    int64
 5   fbs       303 non-null    int64
 6   restecg   303 non-null    int64
 7   thalach   303 non-null    int64
 8   exang     303 non-null    int64
 9   oldpeak   303 non-null    float64
 10  slope     303 non-null    int64
 11  ca        303 non-null    int64
 12  thal      303 non-null    int64
 13  target    303 non-null    int64
dtypes: float64(1), int64(13)
memory usage: 33.3 KB
```

Observations:

- The dataset contains 303 rows and 14 columns.
- All features are numerical values.
- No missing values were present.

Duplicate Removal

One duplicate row was identified and removed from the dataset to prevent bias during model training.

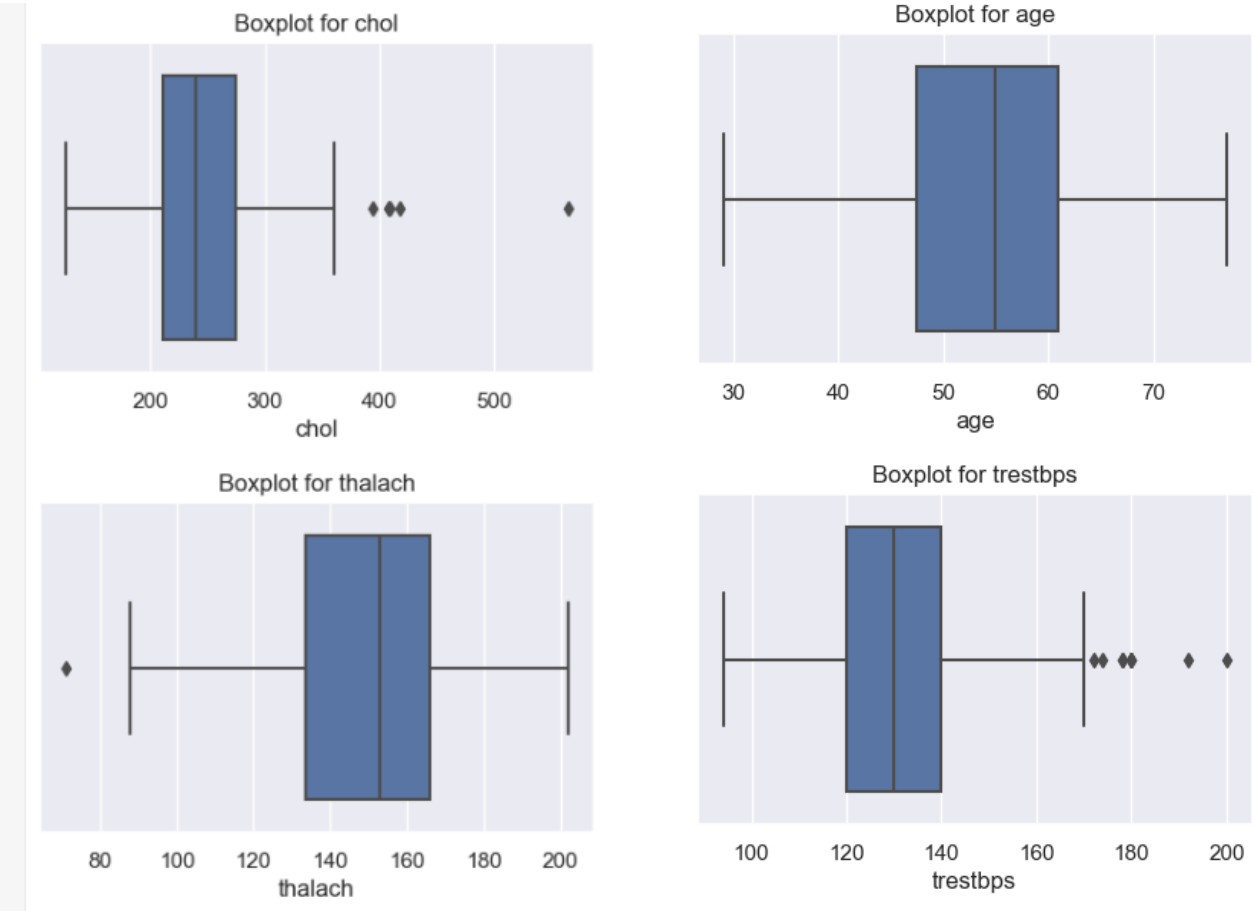
Outlier Detection and Handling

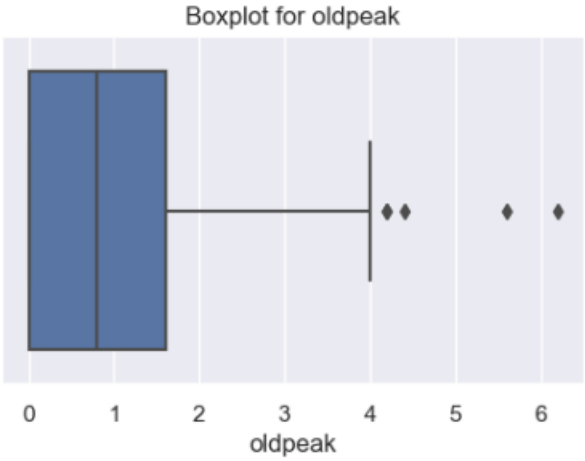
Outliers were detected in several numerical features including:

- resting blood pressure
- cholesterol

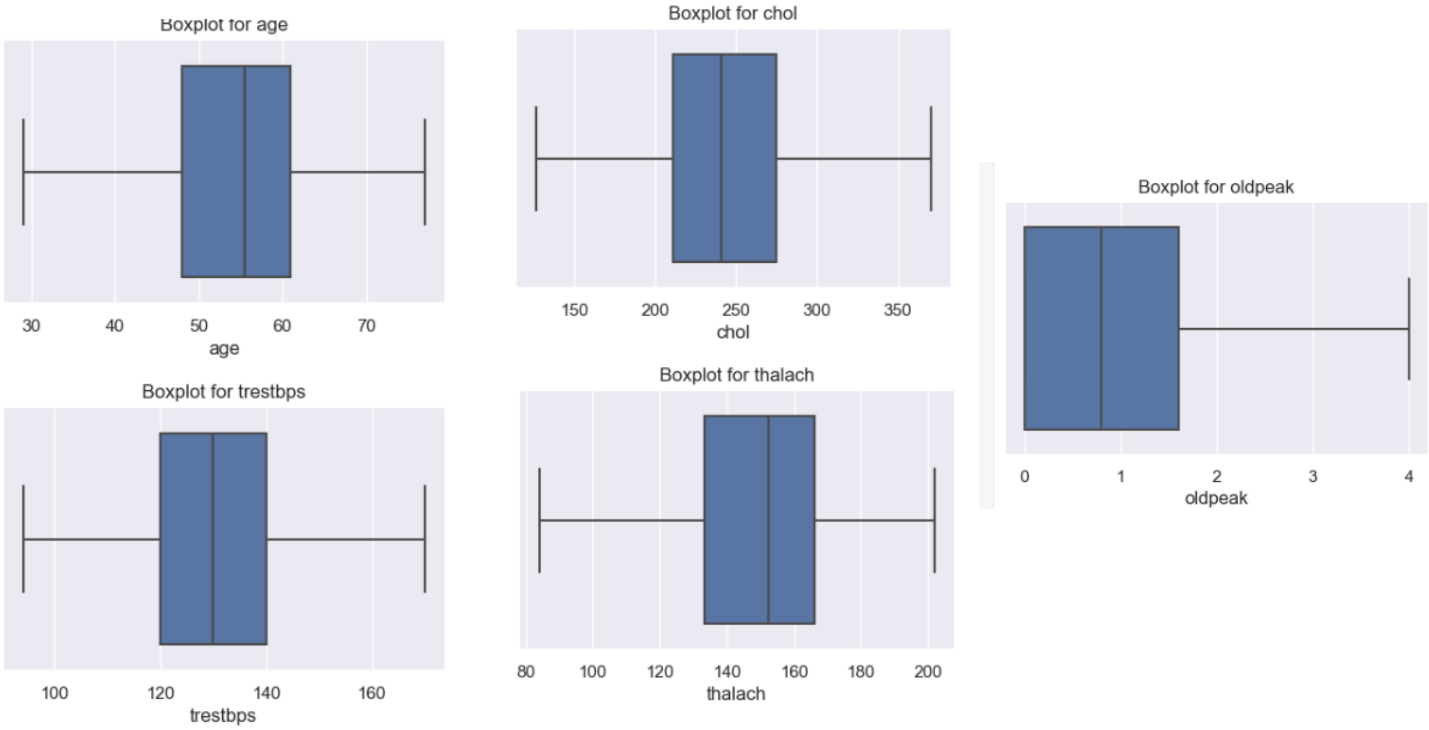
- maximum heart rate
- oldpeak

Outliers were handled using Interquartile Range (IQR) capping.





Boxplots before outlier treatment

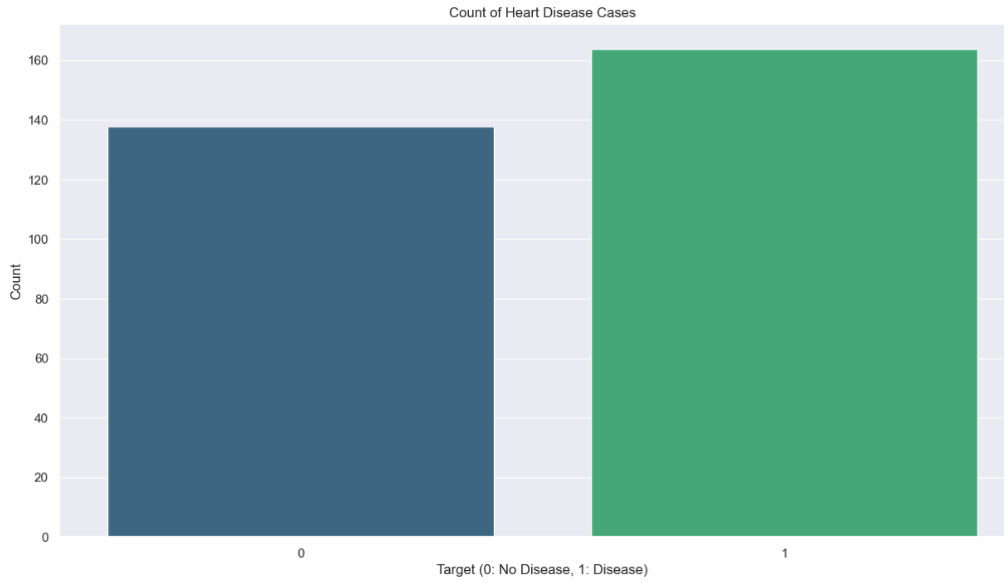


Boxplots after outlier treatment

6. Exploratory Data Analysis (EDA)

EDA was conducted to understand feature distributions and relationships with the target variable.

<u>Distribution of Target Variable</u>

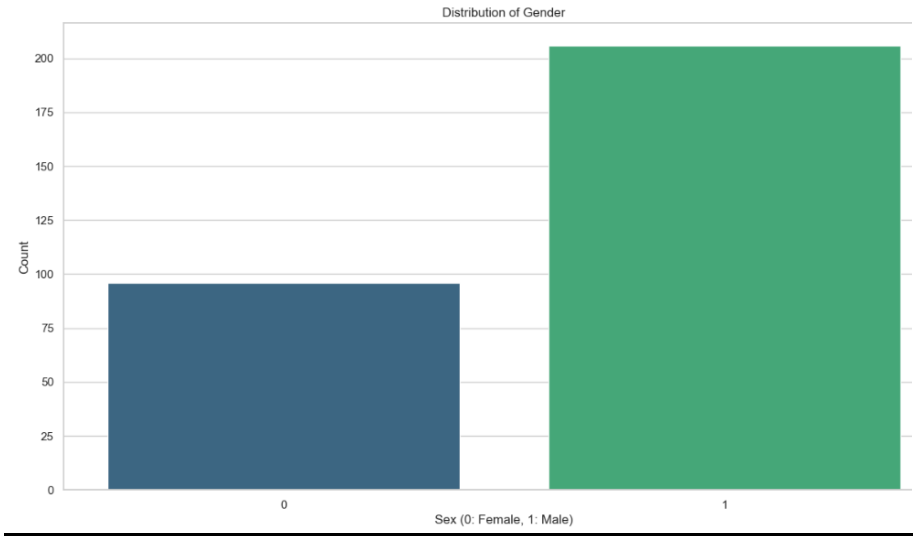


Target distribution countplot

Observation:

The dataset shows a slightly higher number of heart disease cases compared to non-disease cases, though both classes remain relatively balanced.

Gender Distribution

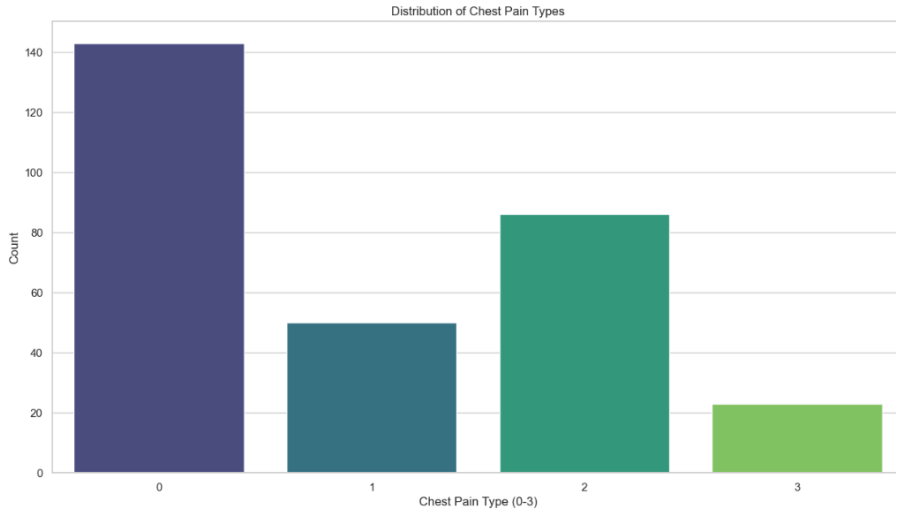


Gender distribution

Observation:

The dataset contains a higher proportion of male patients compared to female patients.

Chest Pain Distribution

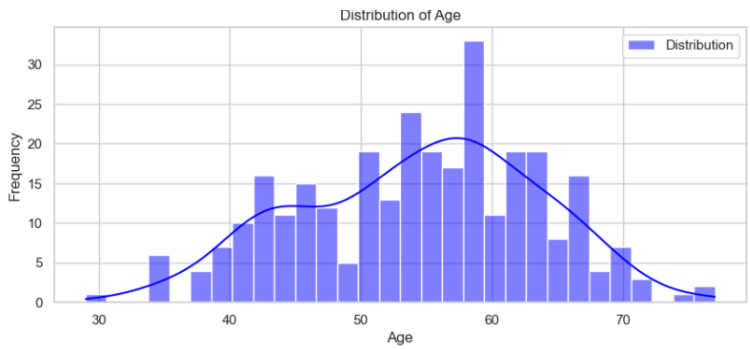


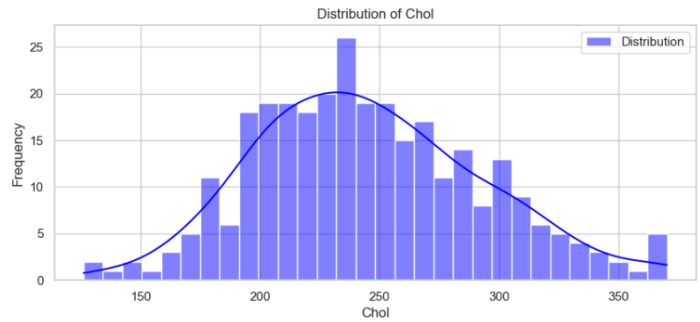
Chest pain type distribution

Observation:

Typical angina and non-anginal pain appear as the most common chest pain types in the dataset.

<u>Continuous Feature Distributions</u>






```
C:\Users\basha\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and
ersion. Convert inf values to NaN before operating instead.
  with pd.option_context('mode.use_inf_as_na', True):
```

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C:\Users\basha\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated
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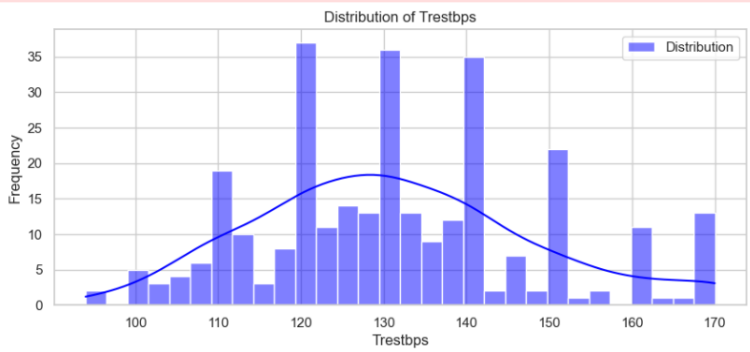
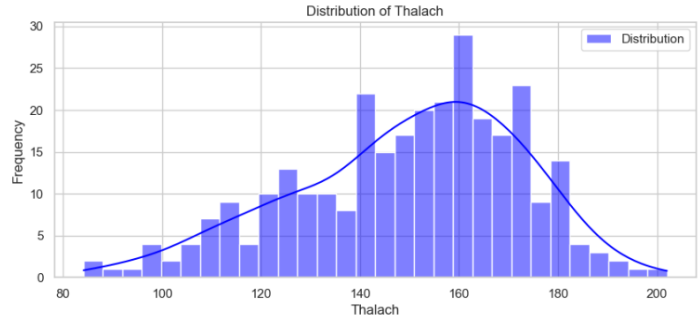
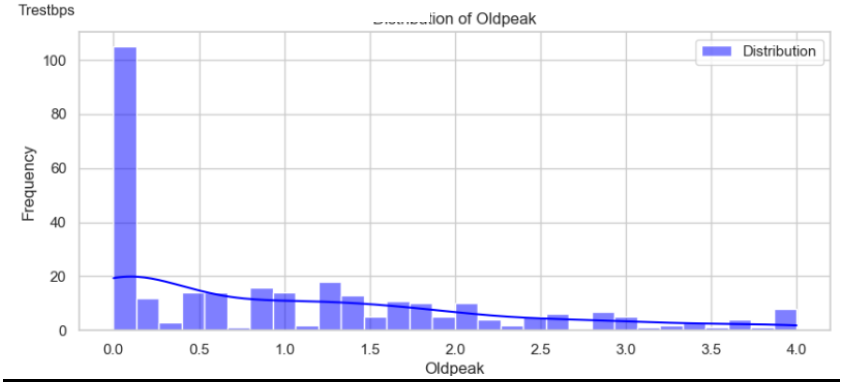




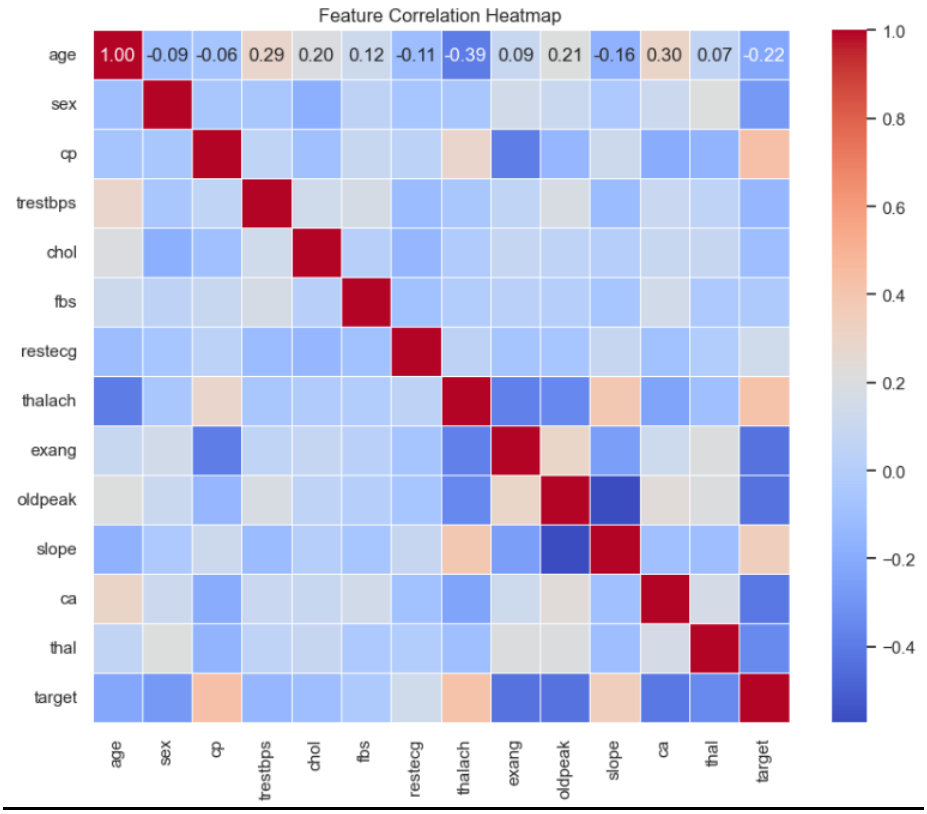


Distribution plots for age, blood pressure, cholesterol, heart rate, oldpeak

Observations:

- Age distribution peaks around 55–60 years.
- Cholesterol values are centered around 200–250 mg/dl.
- Maximum heart rate values typically range between 140–180 bpm.

Feature Correlation Analysis



Correlation heatmap

Observations:

Strong correlations with the target variable include:

- maximum heart rate achieved
- chest pain type
- ST depression
- exercise-induced angina

Weak correlations were observed for:

- fasting blood sugar
- cholesterol
- resting blood pressure

7. Model Development

The dataset was split into training and testing sets using an 80–20 split.

Machine learning models implemented include:

- Logistic Regression
- Naive Bayes
- Support Vector Machine
- K-Nearest Neighbors
- Decision Tree
- Random Forest
- XGBoost

Each model was trained on the training dataset and evaluated on the testing dataset.

8. Model Evaluation

Model performance was evaluated using multiple metrics:

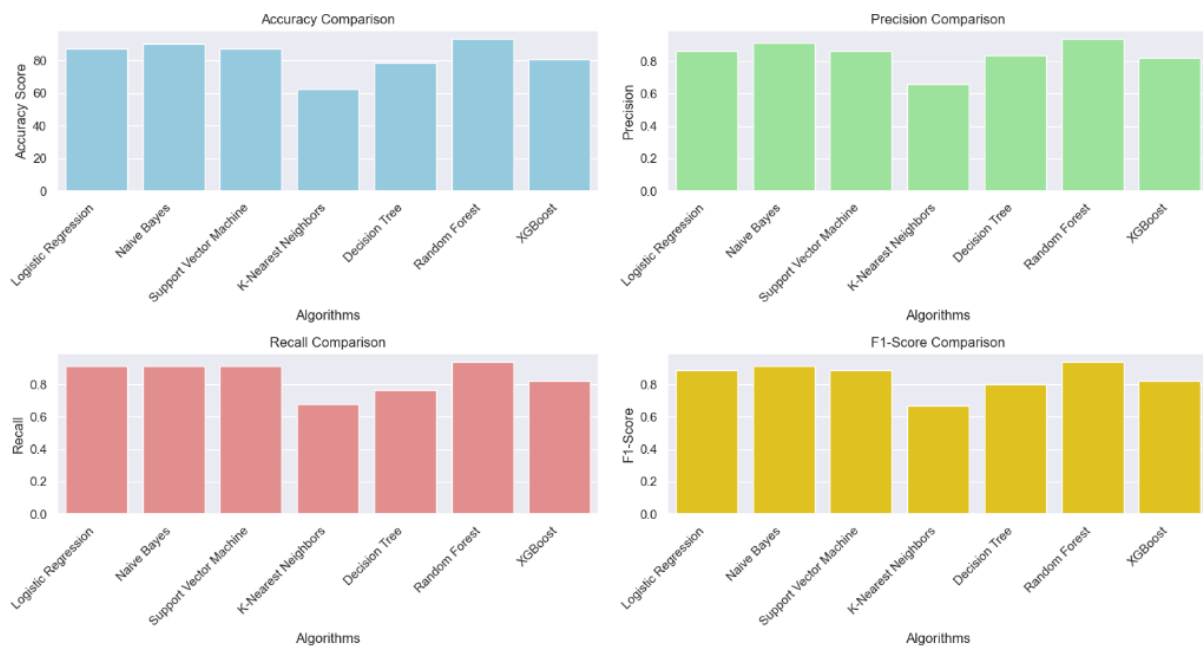
- Accuracy
- Precision
- Recall
- F1 Score
- AUC-ROC

```
Logistic Regression Evaluation Metrics:
Accuracy: 86.89%
Precision: 0.8611111111111112
Recall: 0.9117647058823529
F1-Score: 0.8857142857142858
AUC-ROC: 0.8632897603485838

Naive Bayes Evaluation Metrics:
Accuracy: 90.16%
Precision: 0.9117647058823529
Recall: 0.9117647058823529
F1-Score: 0.9117647058823528
AUC-ROC: 0.9003267973856208

Support Vector Machine Evaluation Metrics:
Accuracy: 86.89%
Precision: 0.8611111111111112
Recall: 0.9117647058823529
F1-Score: 0.8857142857142858
AUC-ROC: 0.8632897603485838

K-Nearest Neighbors Evaluation Metrics:
Accuracy: 62.3%
Precision: 0.6571428571428571
Recall: 0.6764705882352942
F1-Score: 0.6666666666666666
AUC-ROC: 0.6160130718954249
```

```
K-Nearest Neighbors Evaluation Metrics:
Accuracy: 62.3%
Precision: 0.6571428571428571
Recall: 0.6764705882352942
F1-Score: 0.6666666666666666
AUC-ROC: 0.6160130718954249

Decision Tree Evaluation Metrics:
Accuracy: 78.69%
Precision: 0.8387096774193549
Recall: 0.7647058823529411
F1-Score: 0.7999999999999999
AUC-ROC: 0.789760348583878

Random Forest Evaluation Metrics:
Accuracy: 93.44%
Precision: 0.9411764705882353
Recall: 0.9411764705882353
F1-Score: 0.9411764705882353
AUC-ROC: 0.9335511982570807

XGBoost Evaluation Metrics:
Accuracy: 80.33%
Precision: 0.8235294117647058
Recall: 0.8235294117647058
F1-Score: 0.8235294117647058
AUC-ROC: 0.8006535947712419
```

Model Evaluation Metrics

9. Model Comparison

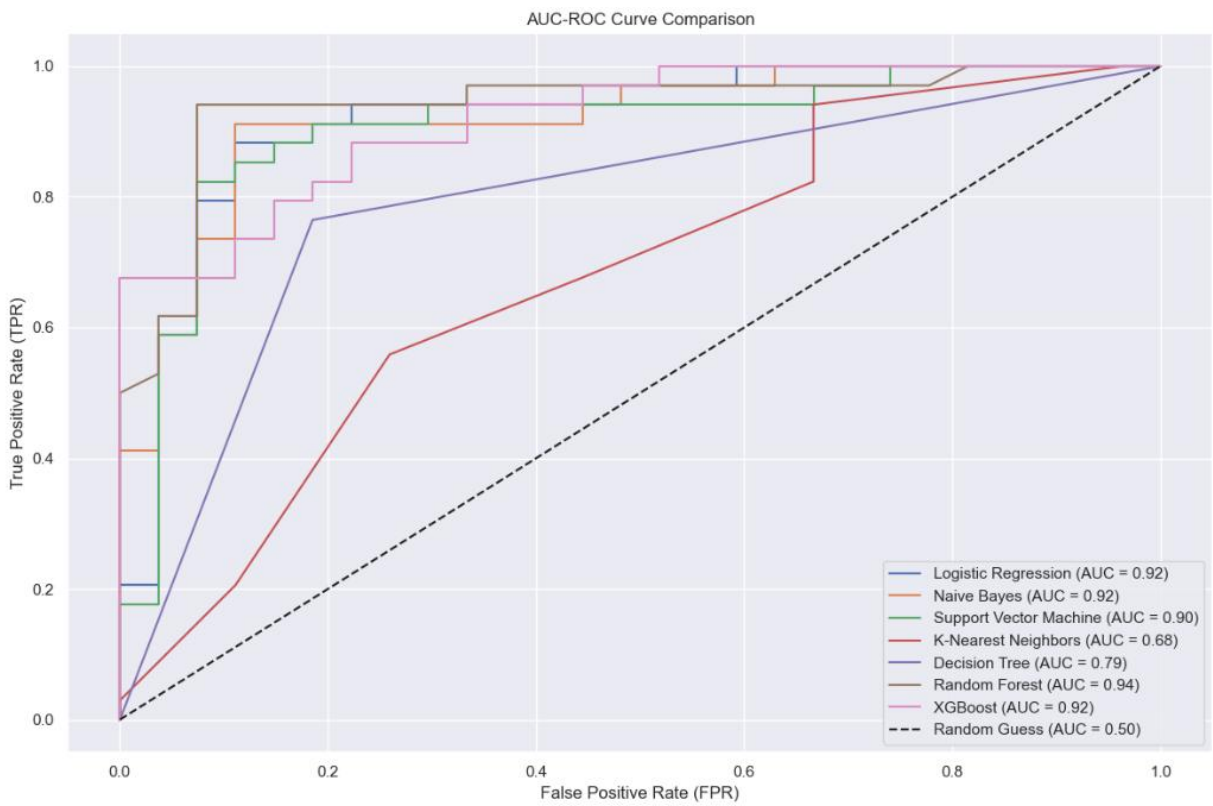


Model comparison bar charts

Observations:

- Random Forest achieved the highest accuracy and balanced performance.
- Naive Bayes and Logistic Regression also performed well.
- KNN showed the lowest performance.

10. ROC Curve Analysis

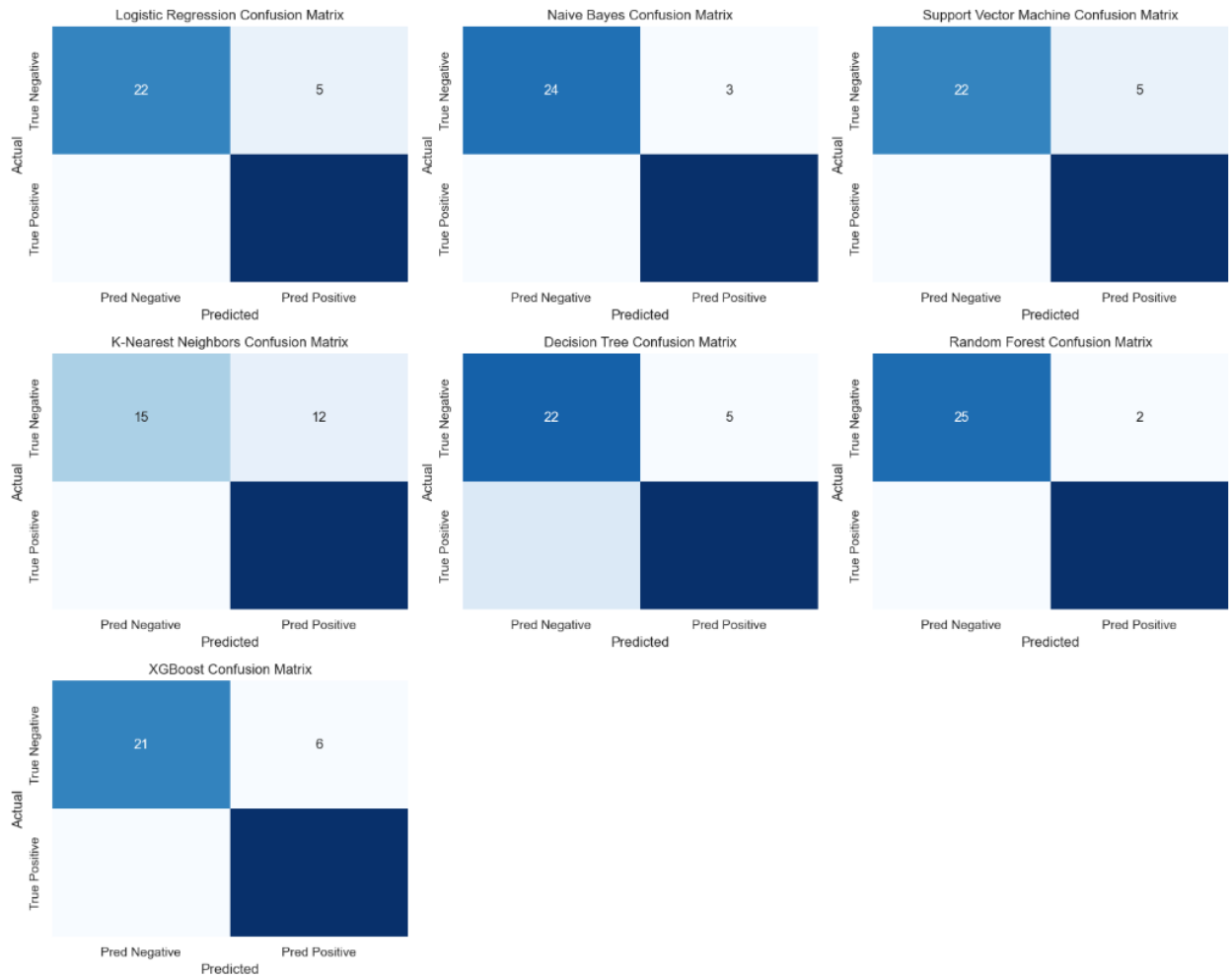


AUC-ROC comparison

Observation:

Random Forest achieved the highest AUC-ROC score, indicating strong class separation capability.

11. Confusion Matrix Analysis



Confusion matrices

Observations:

Random Forest produced the lowest false positives and false negatives, confirming its superior predictive performance.

12. Results and Discussion

The comparative analysis of machine learning models revealed clear performance differences as shown in the following table.

Algorithm	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	86.89	0.861111	0.911765	0.885714	0.863290
Naive Bayes	90.16	0.911765	0.911765	0.911765	0.900327
Support Vector Machine	86.89	0.861111	0.911765	0.885714	0.863290
K-Nearest Neighbors	62.30	0.657143	0.676471	0.666667	0.616013
Decision Tree	78.69	0.838710	0.764706	0.800000	0.789760
Random Forest	93.44	0.941176	0.941176	0.941176	0.933551
XGBoost	80.33	0.823529	0.823529	0.823529	0.800654

Best Performing Model:

- Random Forest emerged as the top-performing model with an accuracy of 93.44%, demonstrating excellent performance across all evaluation metrics. Its high precision (0.941), recall (0.941), F1-score (0.941), and AUC-ROC (0.933) indicate that it is highly reliable in identifying both positive and negative heart disease cases.
- The ensemble nature of Random Forest allows it to capture complex patterns in the data, making it highly suited for this classification task.

Good Performance of Naive Bayes and Logistic Regression:

- Both Naive Bayes (accuracy = 90.16%) and Logistic Regression (accuracy = 86.89%) performed well, with Naive Bayes slightly outperforming Logistic Regression in precision, recall, and AUC-ROC.
- These models are simpler and still managed to perform well, especially Naive Bayes, which balanced precision and recall perfectly, making it a strong contender for this task.

Support Vector Machine (SVM):

- The SVM model (accuracy = 86.89%) showed similar performance to Logistic Regression. While effective in detecting heart disease cases (high recall), it struggled slightly with precision.
- SVM's linear kernel performed well but didn't outperform Random Forest or Naive Bayes in this case.

K-Nearest Neighbors (kNN):

- KNN had the lowest performance with an accuracy of 62.30%, precision of 0.657, and recall of 0.676.
- It struggled with both false positives and false negatives, suggesting that the model's simplicity and lack of tuning may not have been optimal for the dataset.

Decision Tree:

- The Decision Tree model (accuracy = 78.69%) showed decent performance, but it fell behind the more complex models. Its precision and recall values were good, but not as high as those from Random Forest and Naive Bayes.
- This suggests that a more complex or ensemble method like Random Forest could yield better results for heart disease prediction.

XGBoost:

- XGBoost (accuracy = 80.33%) performed moderately, with a balanced precision and recall of 0.824. While effective, it did not surpass Random Forest, Naive Bayes, or Logistic Regression in terms of overall performance.
- However, with further hyperparameter tuning, XGBoost could potentially match or even exceed the performance of Random Forest in certain contexts.

Final thoughts

- Random Forest is the best model for predicting heart disease in this dataset, showing the power of ensemble methods in capturing complex feature relationships.
- Naive Bayes and Logistic Regression are strong contenders for simpler, more interpretable models, with Naive Bayes performing slightly better in balancing precision and recall.
- SVM, Decision Tree, and XGBoost performed well, but the results showed that ensemble models like Random Forest have a distinct advantage in this case.
- KNN showed the weakest performance, which suggests the need for parameter tuning or an alternative algorithm for this type of problem.

13. Conclusion

This project demonstrated the effectiveness of machine learning algorithms for heart disease prediction using clinical data.

Among the models evaluated, Random Forest achieved the highest accuracy and overall performance, making it the most suitable algorithm for this dataset.

The results highlight the potential of machine learning systems as decision-support tools in healthcare applications, particularly for early detection and risk assessment of heart disease.

Future improvements may involve incorporating larger datasets, performing advanced feature engineering, and optimizing hyperparameters to further enhance predictive performance.

14. Future Work

Future improvements could include:

- Hyperparameter tuning for advanced optimization
- Integration of deep learning models
- Incorporation of larger clinical datasets
- Development of a web-based prediction system
- Deployment of the model in healthcare decision-support platforms